

Trần Tấn Phát

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LAB3

Bài làm

Bước 1: Nhập data customer.csv ở lab3

LAB1

New Uploaded Files dataset

/customers.csv 2.65 MB

ADD MORE FILES

Your dataset is almost ready. Here is a preview.

New dataset name: customers [CREATE](#) [CONFIGURE FORMAT](#)

Used /customers.csv (2.65 MB) to parse data. Used format csv and parsed 6 columns

customerID	gender	birthdate	user_agent	ip_address	campaign
886900	M	1953/10/5	Mozilla/5.0 (Windows NT 10.0; Win64; x64; AppleWebKit/537.36 (KHTML, like Gecko) Chrome/51.0.2704.79 Safari/537.36 Edge/14.14393	69.8.136.110	False
038040	F	1944/12/21	Mozilla/5.0 (Windows NT 10.0; WOW64; AppleWebKit/537.36 (KHTML, like Gecko) Chrome/56.0.2924.87 Safari/537.36	67.74.254.174	False
698696	F	1983/12/1	Mozilla/5.0 (Windows NT 6.3; WOW64; AppleWebKit/537.36 (KHTML, like Gecko) Chrome/56.0.2924.87 Safari/537.36	95.135.70.206	False
599696	M	1990/9/23	Mozilla/5.0 (Macintosh; Intel Mac OS X 10_11_6; AppleWebKit/602.3.8 (KHTML, like Gecko) Version/10.0.3 Safari/602.4.6	129.50.205.26	False
496907	M	1869/6/8	Mozilla/5.0 (Windows NT 6.1; Win64; x64; AppleWebKit/537.36 (KHTML, like Gecko) Chrome/56.0.2924.87 Safari/537.36	81.98.234.29	False
486658	F	1948/7/29	Mozilla/5.0 (Windows NT 10.0; Win64; x64; AppleWebKit/537.36 (KHTML, like Gecko) Chrome/56.0.2924.87 Safari/537.36	76.177.15.150	False
148731	F	1880/7/18	Mozilla/5.0 (Windows NT 10.0; Win64; x64; AppleWebKit/537.36 (KHTML, like Gecko) Chrome/56.0.2924.87 Safari/537.36	174.81.130.76	False

Upload to connection: Default (in DSS data dir.)

Bước 2: Dữ liệu sau khi nhập dữ liệu

LAB1

Flow

APPLY A VIEW

SELECT ALL ITEMS 4

+ ADD ITEM

customers

orders

orders_prepared

Visual recipes

- Sync
- Prepare
- Sample / Filter
- Group
- Distinct
- Window
- Join
- Fuzzy join
- Geo join
- Split
- Top N
- Sort
- Pivot
- Stack

Code recipes

- Python
- R
- SQL
- Shell
- Hive
- Impala
- Spark SQL
- PySpark
- Spark R

LLM recipes

- Prompt
- Classify
- Summarize
- Feedback
- Flow tune
- Predict

PREVIEW FLOW ACTIONS

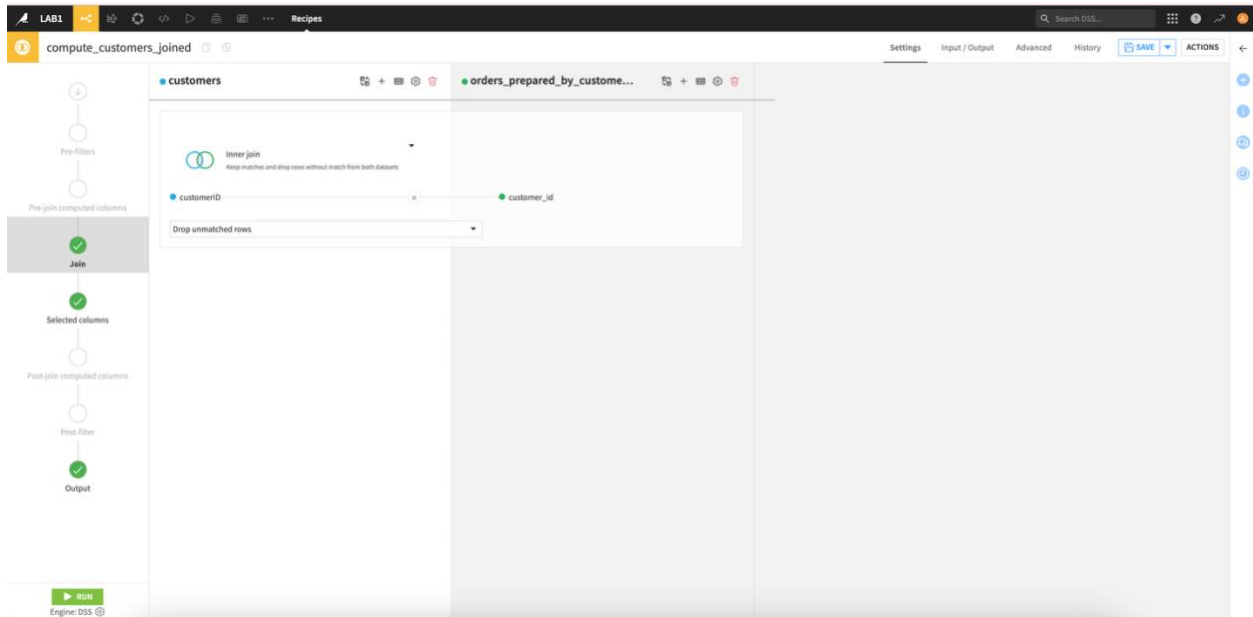
Bước 3:

The screenshot shows the Databricks interface with a 'New join recipe' dialog box open. The dialog box has two main sections: 'Input datasets' and 'Output dataset'. In the 'Input datasets' section, the 'customers' dataset is selected. The 'Output dataset' section is currently empty. A message at the bottom of the dialog states 'This recipe requires two input datasets.' The background shows a table of customer data with columns: customerID, gender, birthdate, and user_agent.

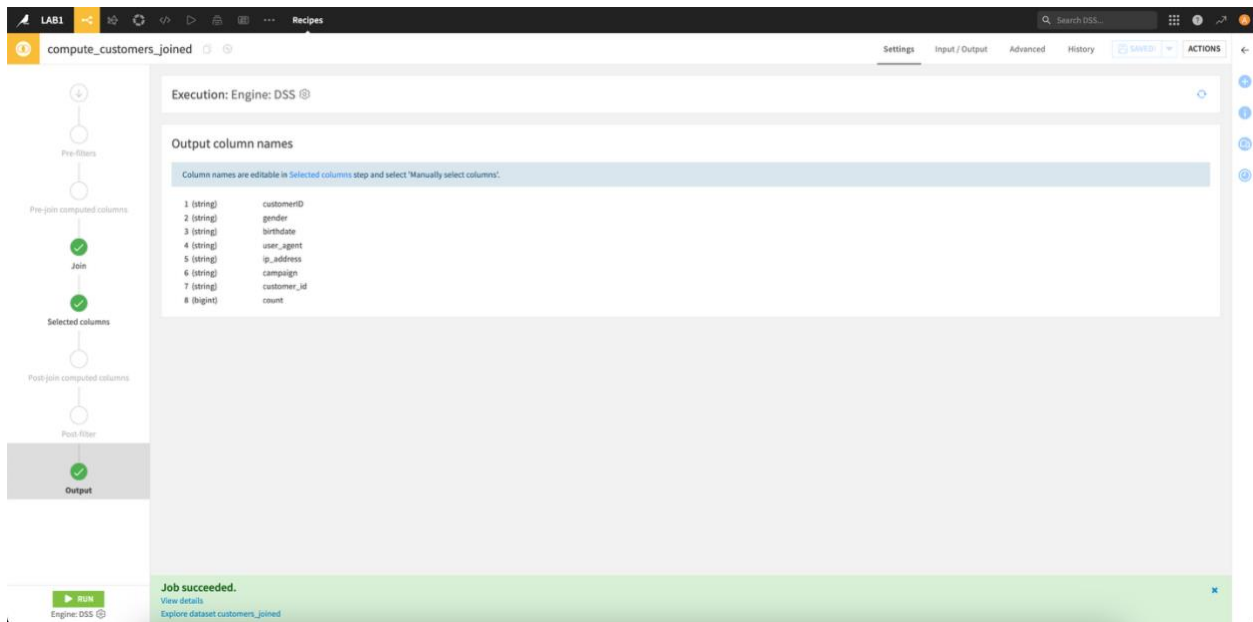
Bước 4:

The screenshot shows the Databricks interface with a 'Left join' recipe configuration. The recipe is named 'compute_customers_joined'. It shows a 'Left join' operation between 'customers' and 'orders_prepared_by_customers'. The join key is 'customerID' from the left dataset and 'customer_id' from the right dataset. The 'Drop unmatched rows' option is selected. The interface also shows a flow diagram on the left and a 'RUN' button at the bottom.

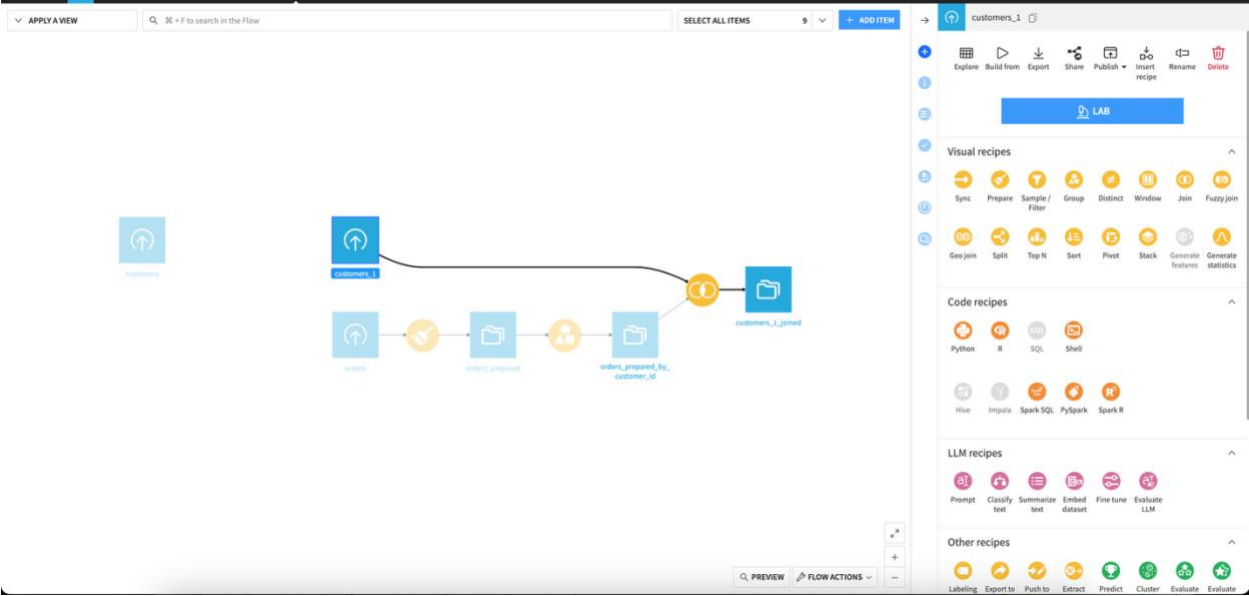
Bước 5: Nhấn đổi lại thành inner join



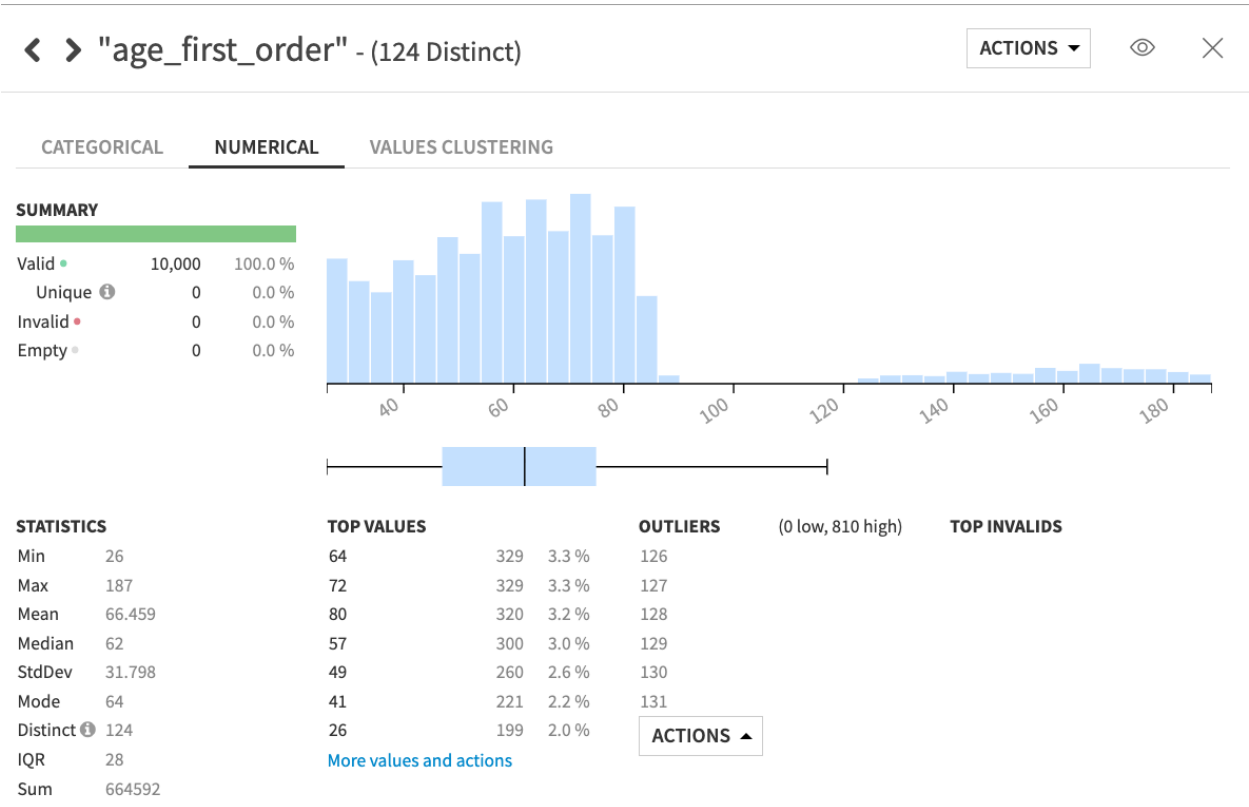
Bước 6: Chạy chương trình

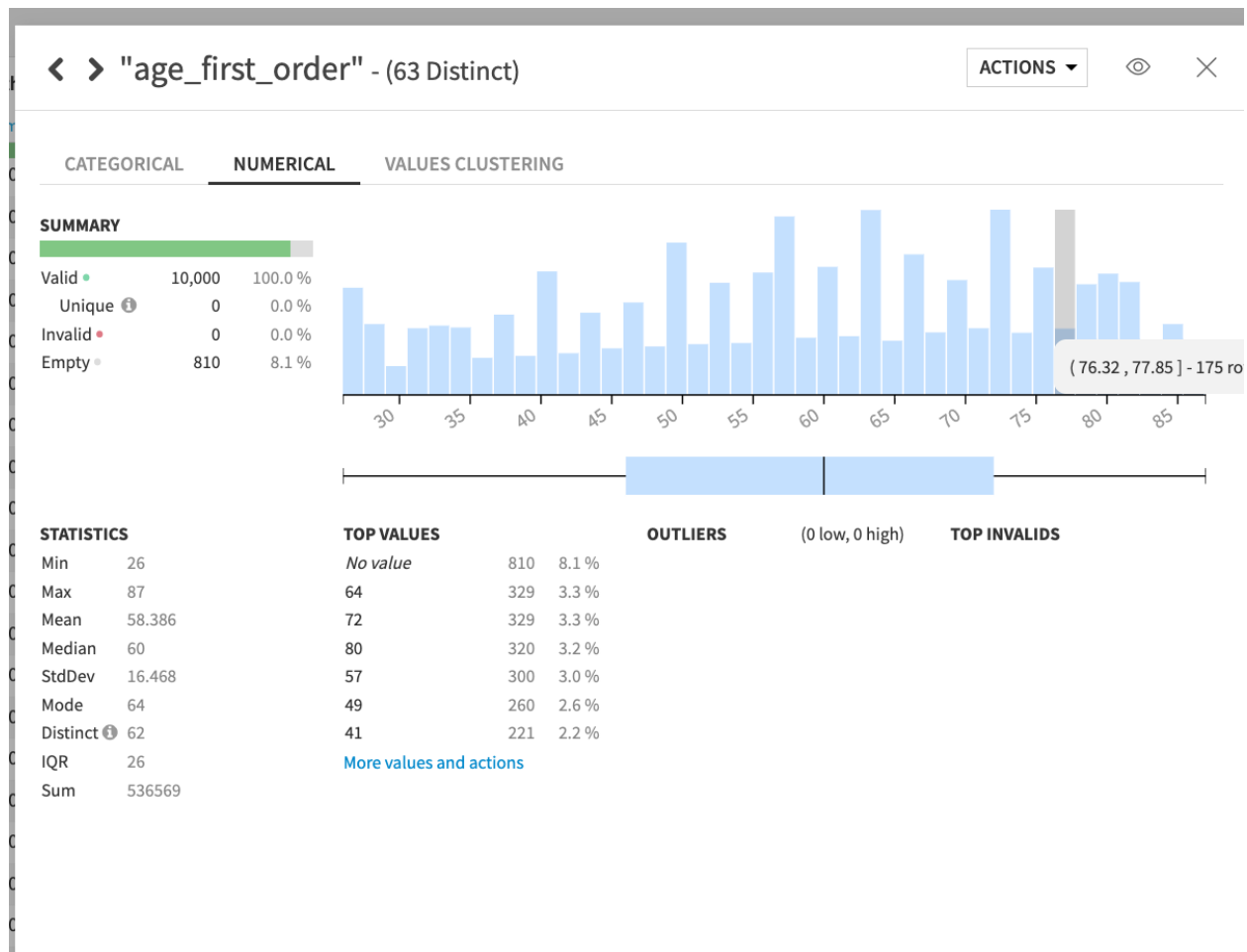


Bước 7: Mở trang customers_order_joined



Bước 8: Phân tích cột age_first_order





Bước 9: Xóa cột birthdate và first_order_date

Analyze customers_1_joined

Script Sample settings Script output on Sample

4 steps First 10,000 rows 10,000 rows (10000)

DISPLAY TABLE COLUMNS   

10,000 rows 8 columns

customerID	gender	age_first_order	user_agent	ip_address	campaign	customer_id	count
Integer	Gender	Integer	User-Agent	IP address	Boolean	Integer	Integer
000018	F	71	Mozilla/5.0 (Windows NT 10.0; WOW64) AppleWebKit/...	46.168.91.127	False	000018	3
000071	F	59	Mozilla/5.0 (Macintosh; Intel Mac OS X 10_12_3) Ap...	106.140.104.103	True	000071	3
000440	M	55	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWe...	45.209.292.83	True	000440	5
000452	M	73	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWe...	150.35.247.187	False	000452	2
000652	M	32	Mozilla/5.0 (Windows NT 6.1; WOW64; rv:51.0) Geck...	48.58.46.246	False	000652	2
000747	M	61	Mozilla/5.0 (Windows NT 6.1; WOW64) AppleWebKit/...	40.35.78.53	False	000747	2
000759	M	36	Mozilla/5.0 (Macintosh; Intel Mac OS X 10_11_6) Ap...	180.227.194.42	True	000759	3
000759	M		Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:51.0) Ge...	48.21.157.116	False	000759	3
000879	F	69	Mozilla/5.0 (Macintosh; Intel Mac OS X 10_12_3) Ap...	52.76.90.175	False	000879	3
000887	F	37	Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.3...	176.125.60.83	True	000887	2
001226	F	71	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWe...	175.49.47.202	False	001226	3
001247	F	69	Mozilla/5.0 (Macintosh; Intel Mac OS X 10_12_3) Ap...	63.33.211.5	True	001247	4
001275	F	77	Mozilla/5.0 (Windows NT 6.1; Win64; x64) AppleWe...	54.61.83.179	False	001275	4
001411	F	80	Mozilla/5.0 (Windows NT 6.1; WOW64; rv:51.0) Geck...	101.179.50.217	True	001411	6
001492	F	35	Mozilla/5.0 (Windows NT 10.0; WOW64) AppleWebKit/...	133.92.58.240	True	001492	7
001571	M	53	Mozilla/5.0 (Macintosh; Intel Mac OS X 10_12_3) Ap...	120.97.110.47	False	001571	3
001682	M	80	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWe...	120.59.101.195	False	001682	1
001771	M	26	Mozilla/5.0 (Macintosh; Intel Mac OS X 10_12_3) Ap...	125.75.27.154	False	001771	1
001771	F	80	Mozilla/5.0 (Macintosh; Intel Mac OS X 10_11_6) Ap...	100.73.182.67	False	001771	1
002158	M	79	Mozilla/5.0 (Windows NT 10.0; WOW64; rv:51.0) Geck...	49.197.152.84	False	002158	1
002241		75	Mozilla/5.0 (Macintosh; Intel Mac OS X 10_12_3) Ap...	151.153.5.130	True	002241	10
002402		46	Mozilla/5.0 (Macintosh; Intel Mac OS X 10_12_3) Ap...	166.2.135.170	False	002402	1
002431	M	57	Mozilla/5.0 (Windows NT 10.0; WOW64) AppleWebKit/...	60.54.76.30	False	002431	2
002453	F	56	Mozilla/5.0 (Macintosh; Intel Mac OS X 10_11_6) Ap...	43.2.171.119	True	002453	3
002555	M		Mozilla/5.0 (Macintosh; Intel Mac OS X 10_11_6) Ap...	113.196.23.53	False	002555	1

Bước 10: Xử lý cột ip_address và user_agent

Parse date in birthdate

10000

Compute time difference between birthdate and now

10000

Clear values outside [34,100] in age_first_order

818

Remove birthdate

10000

Classify User-Agent in user_agent

10000

Column

user_agent

+ ADD A NEW STEP

+ ADD A GROUP

customerID	gender	age_first_order	user_agent	user_agent_type	user_agent_category	user_agent_brand	user_agent_version	user_agent_os	user_agent_osversion	user_agent
Integer	Gender	Integer	User-Agent	Text	Text	Text	Text	Text	Text	Text
000018	F	71	Mozilla/5.0 (Windows NT 10.0; WOW64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/66.0.3359.139 Safari/537.36	browser	Chrome	Chrome	56.0.2924.87	Windows	Windows 10	64 bits
000071	F	59	Mozilla/5.0 (Macintosh; Intel Mac OS X 10_12_3) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/66.0.3359.139 Safari/537.36	browser	Chrome	Chrome	56.0.2924.87	MacOS X	10.12.3	
000440	M	55	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/66.0.3359.139 Safari/537.36	browser	Chrome	Chrome	56.0.2924.87	Windows	Windows 10	64 bits
000452	M	73	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/66.0.3359.139 Safari/537.36	browser	Chrome	Chrome	56.0.2924.87	Windows	Windows 10	64 bits
000652	M	32	Mozilla/5.0 (Windows NT 6.1; WOW64; rv:51.0) Gecko/20100101 Firefox/51.0	browser	Firefox (=v4)	Firefox	51.0	Windows	Windows 7	32 bits
000747	M	61	Mozilla/5.0 (Windows NT 6.1; WOW64; rv:51.0) Gecko/20100101 Firefox/51.0	browser	Chrome	Chrome	56.0.2924.87	Windows	Windows 7	32 bits
000759	M	36	Mozilla/5.0 (Macintosh; Intel Mac OS X 10_11_6) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/66.0.3359.139 Safari/537.36	browser	Chrome	Chrome	56.0.2924.87	MacOS X	10.11.6	
000759	M		Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:51.0) Gecko/20100101 Firefox/51.0	browser	Firefox (=v4)	Firefox	51.0	Linux		Ubuntu
000879	F	69	Mozilla/5.0 (Macintosh; Intel Mac OS X 10_12_3) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/66.0.3359.139 Safari/537.36	browser	Chrome	Chrome	56.0.2924.87	MacOS X	10.12.3	
000887	F	37	Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/66.0.3359.139 Safari/537.36	browser	Chrome	Chrome	56.0.2924.87	Linux		
001226	F	71	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/66.0.3359.139 Safari/537.36	browser	Chrome	Chrome	56.0.2924.87	Windows	Windows 10	64 bits
001247	F	69	Mozilla/5.0 (Macintosh; Intel Mac OS X 10_12_3) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/66.0.3359.139 Safari/537.36	browser	Safari	Safari	10.0.3	MacOS X	10.12.3	
001275	F	77	Mozilla/5.0 (Windows NT 6.1; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/66.0.3359.139 Safari/537.36	browser	Chrome	Chrome	56.0.2924.87	Windows	Windows 7	32 bits
001411		80	Mozilla/5.0 (Windows NT 6.1; WOW64; rv:51.0) Gecko/20100101 Firefox/51.0	browser	Firefox (=v4)	Firefox	51.0	Windows	Windows 7	32 bits
001492	F	35	Mozilla/5.0 (Windows NT 10.0; WOW64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/66.0.3359.139 Safari/537.36	browser	Chrome	Chrome	56.0.2924.87	Windows	Windows 10	64 bits
001571	M	53	Mozilla/5.0 (Macintosh; Intel Mac OS X 10_12_3) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/66.0.3359.139 Safari/537.36	browser	browser	Chrome	56.0.2924.87	MacOS X	10.12.3	
001682	M	80	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/66.0.3359.139 Safari/537.36	browser	Chrome	Chrome	56.0.2924.87	Windows	Windows 10	64 bits
001771	M	26	Mozilla/5.0 (Macintosh; Intel Mac OS X 10_11_6) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/66.0.3359.139 Safari/537.36	browser	Chrome	Chrome	56.0.2924.87	MacOS X	10.12.3	
001819	F	80	Mozilla/5.0 (Macintosh; Intel Mac OS X 10_11_6) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/66.0.3359.139 Safari/537.36	browser	Chrome	Chrome	56.0.2924.87	MacOS X	10.11.6	
002158	M	79	Mozilla/5.0 (Windows NT 10.0; WOW64; rv:51.0) Gecko/20100101 Firefox/51.0	browser	Firefox (=v4)	Firefox	51.0	Windows	Windows 10	64 bits
002241		75	Mozilla/5.0 (Macintosh; Intel Mac OS X 10_12_3) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/66.0.3359.139 Safari/537.36	browser	Chrome	Chrome	56.0.2924.87	MacOS X	10.12.3	
002402		46	Mozilla/5.0 (Macintosh; Intel Mac OS X 10_12_3) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/66.0.3359.139 Safari/537.36	browser	Safari	Safari	10.0.3	MacOS X	10.12.3	
002431	M	57	Mozilla/5.0 (Macintosh; Intel Mac OS X 10_12_3) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/66.0.3359.139 Safari/537.36	browser	Chrome	Chrome	56.0.2924.87	Windows	Windows 10	64 bits
002453	F	56	Mozilla/5.0 (Macintosh; Intel Mac OS X 10_10_5) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/66.0.3359.139 Safari/537.36	browser	Chrome	Chrome	56.0.2924.87	MacOS X	10.10.5	
002555	M		Mozilla/5.0 (Macintosh; Intel Mac OS X 10_11_6) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/66.0.3359.139 Safari/537.36	browser	Chrome	Chrome	56.0.2924.87	MacOS X	10.11.6	
002938	M	51	Mozilla/5.0 (Macintosh; Intel Mac OS X 10_11_6) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/66.0.3359.139 Safari/537.36	browser	Safari	Safari	10.0.3	MacOS X	10.11.6	

Geo-locate IP in ip_address

8879

IP Column

ip_address

Prefix for generated output columns

ip_address_

☒ Extract country

☐ Extract country code

☐ Extract country code (3 letters)

☐ Extract region

☐ Extract city

☐ Extract postal code

☐ Extract lat/long

☒ Extract GeoPoint

☐ Extract timezone

☐ Extract continent code

+ ADD A NEW STEP

+ ADD A GROUP

customerID	gender	birthdate	user_agent	ip_address	ip_address_geo_point	ip_address_country	campaign	customer_id	count
Integer	Gender	Date (timestamp)	User-Agent	IP address	GeoPoint	Country	Boolean	Integer	Integer
000018	F	1954/8/19	Mozilla/5.0 (Windows NT 10.0; WOW64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/66.0.3359.139 Safari/537.36	46.168.91.127	POINT(52.2394 21.632)	Poland	False	000018	3
000071	F	1966/12/20	Mozilla/5.0 (Macintosh; Intel Mac OS X 10_12_3) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/66.0.3359.139 Safari/537.36	106.140.104.103	POINT(35.6897 139.28)	Japan	True	000071	3
000440	M	1970/6/24	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/66.0.3359.139 Safari/537.36	45.209.202.83	POINT(2 2)	Ghana	True	000440	5
000452	M	1952/11/8	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/66.0.3359.139 Safari/537.36	150.35.247.187	POINT(35.6897 139.28)	Japan	False	000452	2
000652	M	1993/11/6	Mozilla/5.0 (Windows NT 6.1; WOW64; rv:51.0) Gecko/20100101 Firefox/51.0	48.58.46.246	POINT(37.751 97.822)	United States	False	000652	2
000747	M	1964/11/5	Mozilla/5.0 (Windows NT 6.1; WOW64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/66.0.3359.139 Safari/537.36	40.35.78.53	POINT(37.751 97.822)	United States	False	000747	2
000759	M	1988/4/3	Mozilla/5.0 (Macintosh; Intel Mac OS X 10_11_6) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/66.0.3359.139 Safari/537.36	180.227.194.42	POINT(37.1548 127.0674)	South Korea	True	000759	3
000759	M	1859/2/5	Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:51.0) Gecko/20100101 Firefox/51.0	48.21.157.116	POINT(37.751 97.822)	United States	False	000759	3
000879	F	1956/2/6	Mozilla/5.0 (Macintosh; Intel Mac OS X 10_12_3) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/66.0.3359.139 Safari/537.36	52.76.90.175	POINT(1.3036 103.8554)	Singapore	False	000879	3
000887	F	1988/10/17	Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/66.0.3359.139 Safari/537.36	176.129.60.83	POINT(46.1963 18.24)	France	True	000887	2
001226	F	1954/3/29	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/66.0.3359.139 Safari/537.36	175.49.47.202	POINT(34.7732 113.722)	China	False	001226	3
001247	F	1956/12/13	Mozilla/5.0 (Macintosh; Intel Mac OS X 10_12_3) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/66.0.3359.139 Safari/537.36	63.33.211.5	POINT(53.3382 6.2591)	Ireland	True	001247	4
001275	F	1948/5/26	Mozilla/5.0 (Windows NT 6.1; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/66.0.3359.139 Safari/537.36	54.61.83.179	POINT(37.751 97.822)	United States	False	001275	4
001411		1945/9/28	Mozilla/5.0 (Windows NT 6.1; WOW64; rv:51.0) Gecko/20100101 Firefox/51.0	101.179.50.217	POINT(34.9274 138.6005)	Australia	True	001411	6
001492	F	1990/6/24	Mozilla/5.0 (Windows NT 10.0; WOW64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/66.0.3359.139 Safari/537.36	133.92.58.240	POINT(34.3273 134.044)	Japan	True	001492	7
001571	M	1972/2/16	Mozilla/5.0 (Macintosh; Intel Mac OS X 10_12_3) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/66.0.3359.139 Safari/537.36	120.97.110.47	POINT(23.5 121.221)	Taiwan	False	001571	3
001682	M	1945/2/10	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/66.0.3359.139 Safari/537.36	120.59.101.195	POINT(28.6328 77.2204)	India	False	001682	1
001771	M	1999/2/20	Mozilla/5.0 (Macintosh; Intel Mac OS X 10_12_3) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/66.0.3359.139 Safari/537.36	125.75.27.154	POINT(36.0647 103.8397)	China	False	001771	1
002158	F	1946/10/18	Mozilla/5.0 (Macintosh; Intel Mac OS X 10_11_6) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/66.0.3359.139 Safari/537.36	100.73.182.67			False	002158	1
002241	M	1946/8/17	Mozilla/5.0 (Windows NT 10.0; WOW64; rv:51.0) Gecko/20100101 Firefox/51.0	49.197.152.84	POINT(47.4743 153.0224)	Australia	False	002241	10
002402		1950/6/3	Mozilla/5.0 (Macintosh; Intel Mac OS X 10_12_3) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/66.0.3359.139 Safari/537.36	151.153.5.130	POINT(1.3673 103.8014)	Singapore	True	002402	1
002431	M	1979/6/11	Mozilla/5.0 (Macintosh; Intel Mac OS X 10_12_3) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/66.0.3359.139 Safari/537.36	166.2.135.170	POINT(37.751 97.822)	United States	False	002431	1
002453	F	1968/5/28	Mozilla/5.0 (Windows NT 10.0; WOW64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/66.0.3359.139 Safari/537.36	60.54.76.30	POINT(3.7145 101.9328)	Malaysia	False	002453	2
002555	M	1968/5/28	Mozilla/5.0 (Macintosh; Intel Mac OS X 10_10_5) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/66.0.3359.139 Safari/537.36	43.2.171.119	POINT(3.7145 103.8014)	Singapore	True	002555	3
002555	M	1861/6/22	Mozilla/5.0 (Macintosh; Intel Mac OS X 10_11_6) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/66.0.3359.139 Safari/537.36	113.196.23.53	POINT(23.5 121.221)	Taiwan	False	002555	1
002938	M	1997/4/21	Mozilla/5.0 (Macintosh; Intel Mac OS X 10_11_6) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/66.0.3359.139 Safari/537.36	161.196.100.50	POINT(6.8 107.66)	Venezuela	False	002938	1

Script Sample settings Script output on Sample

4 steps First 10,000 rows 10,000 rows (of 10000)

Geo-locate IP in ip_address
Remove ip_address
Classify User-Agent in user_agent
Remove 5 columns
Column: user_agent_version, user_agent_osversion, user_agent_type, user_agent_category, user_agent
+ ADD A NEW STEP
+ ADD A GROUP

customerID	gender	birthdate	user_agent_brand	user_agent_os	user_agent_osflavor	ip_address_geopoint	ip_address_country	campaign	customer_id	count
integer	Gender	Date (unparsed)	Text	Text	Text	GeoPoint	Country	Boolean	integer	integer
000018	F	1954/8/19	Chrome	Windows	Meaning (inferred from data)	POINT(21.0362 52.2394)	Poland	False	000018	3
000071	F	1966/12/20	Chrome	MacOS X		POINT(139.6895 35.6897)	Japan	True	000071	3
000440	M	1970/6/24	Chrome	Windows		POINT(1.2 8)	Ghana	True	000440	5
000452	M	1952/11/8	Chrome	Windows		POINT(139.6895 35.6897)	Japan	False	000452	2
000652	M	1993/11/6	Firefox	Windows		POINT(97.822 37.751)	United States	False	000652	2
000747	M	1964/11/5	Chrome	Windows		POINT(97.822 37.751)	United States	False	000747	2
000759	M	1989/4/3	Chrome	MacOS X		POINT(127.0674 37.1548)	South Korea	True	000759	3
000759	M	1859/2/5	Firefox	Linux	Ubuntu	POINT(97.822 37.751)	United States	False	000759	3
000879	F	1956/2/6	Chrome	MacOS X		POINT(103.8554 1.3636)	Singapore	False	000879	3
000887	F	1988/10/17	Chrome	Linux		POINT(8.24 46.1363)	France	True	000887	2
001226	F	1954/3/29	Chrome	Windows		POINT(113.722 34.7732)	China	False	001226	3
001247	F	1956/12/13	Safari	MacOS X		POINT(4.2591 53.3382)	Ireland	True	001247	4
001275	F	1948/5/26	Chrome	Windows		POINT(97.822 37.751)	United States	False	001275	4
001411		1945/9/28	Firefox	Windows		POINT(138.6005 -34.9274)	Australia	True	001411	6
001492	F	1990/6/24	Chrome	Windows		POINT(134.044 34.3273)	Japan	True	001492	7
001571	M	1972/2/16	Chrome	MacOS X		POINT(121.23 5)	Taiwan	False	001571	3
001682	M	1945/2/10	Chrome	Windows		POINT(77.2204 28.6328)	India	False	001682	1
001771	M	1999/2/20	Chrome	MacOS X		POINT(103.8397 36.0647)	China	False	001771	1
002101	F	1945/10/18	Chrome	MacOS X			False		002101	1
002158	M	1946/8/17	Firefox	Windows		POINT(153.0224 -27.4743)	Australia	False	002158	1
002241		1950/6/3	Chrome	MacOS X		POINT(103.8014 1.3673)	Singapore	True	002241	10
002402		1979/6/11	Safari	MacOS X		POINT(97.822 37.751)	United States	False	002402	1
002431	M	1968/2/22	Chrome	Windows		POINT(101.9328 2.7145)	Malaysia	False	002431	2
002453	F	1969/5/28	Chrome	MacOS X		POINT(103.8014 1.3673)	Singapore	True	002453	3
002555	M	1861/6/22	Chrome	MacOS X		POINT(121.23 5)	Taiwan	False	002555	1
002938	M	1982/4/21	Safari	MacOS X		POINT(146.8)	Vietnam	False	002938	1

Bước 11: Vào add new step nhấn chọn Formula và nhập dòng như trên

Analyze customers_1_joined

Script Sample settings

5 steps First 10,000 rows

Geo-locate IP in ip_address
Remove ip_address
Classify User-Agent in user_agent
Remove 5 columns
Formula (invalid - no desc)
Missing parameter: Formula

Output column: formula_result
Expression
Error column

+ ADD A NEW STEP
+ ADD A GROUP

Formula for formula_result

if(total_sum > 300, "True", "False")

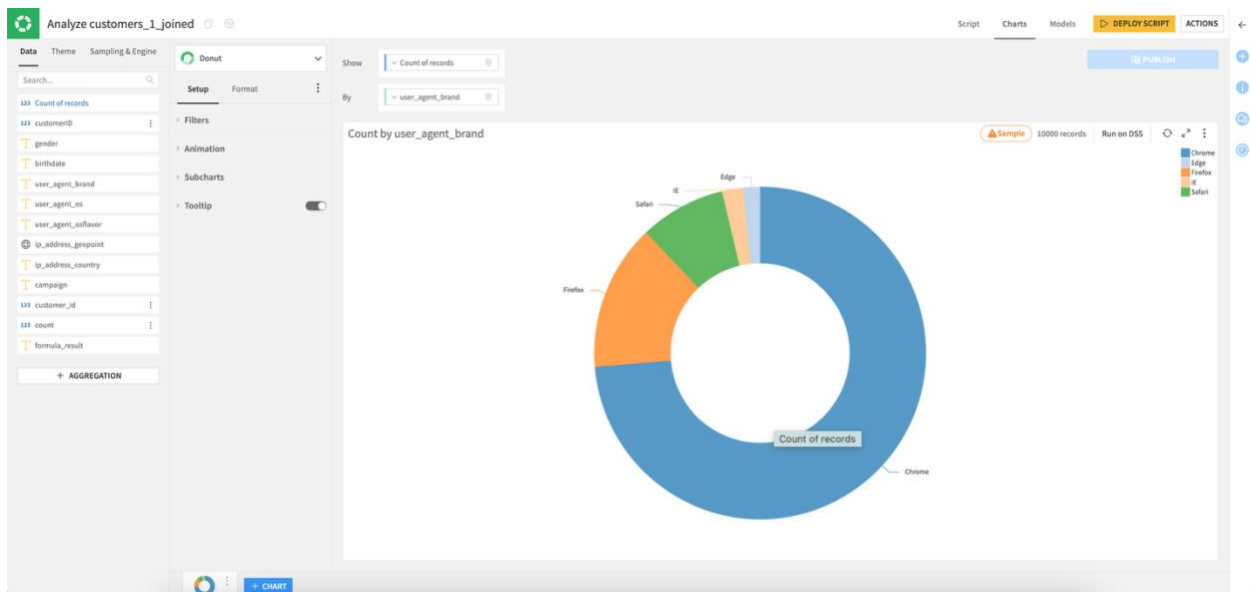
Preview

Sample output	total_sum
False	
False	
False	
False	
False	
False	
False	

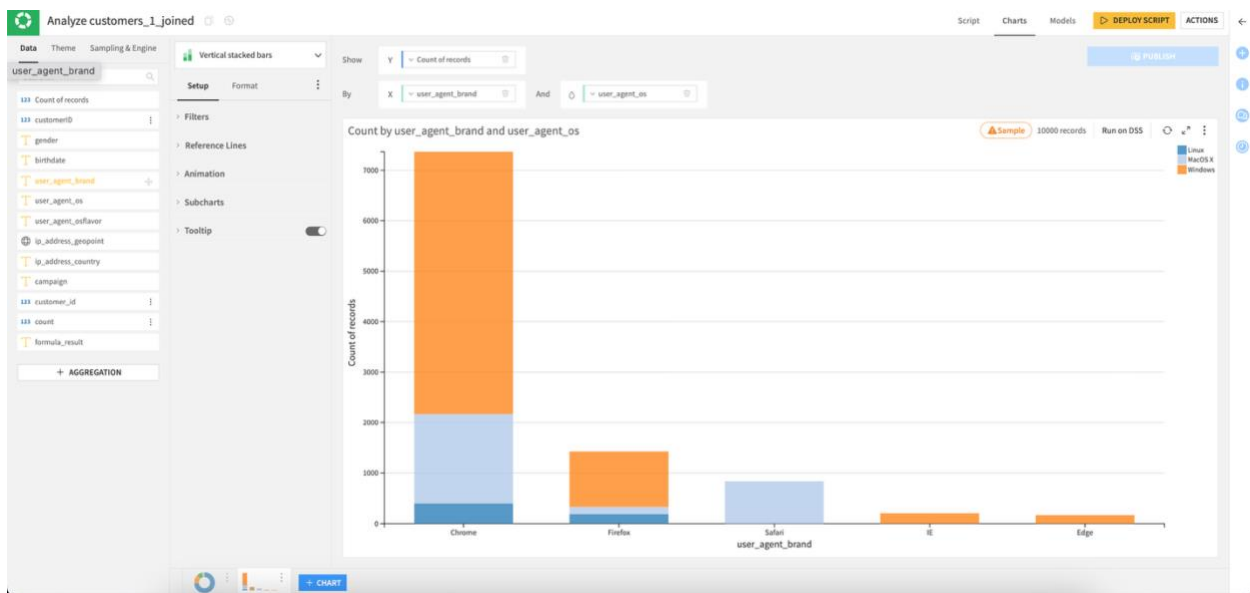
Formula is valid

customerID	gender	birthdate	user_agent_brand	user_agent_os	user_agent_osflavor	ip_address_geopoint	ip_address_country	campaign	customer_id	count
integer	Gender	Date (unparsed)	Text	Text	Text	GeoPoint	Country	Boolean	integer	integer
000018	F	1954/8/19	Chrome	Windows	64 bits	POINT(21.0362 52.2394)	Poland	False	000018	3
000071	F	1966/12/20	Chrome	MacOS X		POINT(139.6895 35.6897)	Japan	True	000071	3
000440	M	1970/6/24	Chrome	Windows	64 bits	POINT(1.2 8)	Ghana	True	000440	5
000452	M	1952/11/8	Chrome	Windows	64 bits	POINT(139.6895 35.6897)	Japan	False	000452	2
000652	M	1993/11/6	Firefox	Windows	32 bits	POINT(97.822 37.751)	United States	False	000652	2
000747	M	1964/11/5	Chrome	Windows	32 bits	POINT(97.822 37.751)	United States	False	000747	2
000759	M	1989/4/3	Chrome	MacOS X		POINT(127.0674 37.1548)	South Korea	True	000759	3
000759	M	1859/2/5	Firefox	Linux	Ubuntu	POINT(97.822 37.751)	United States	False	000759	3
000879	F	1956/2/6	Chrome	MacOS X		POINT(103.8554 1.3636)	Singapore	False	000879	3
000887	F	1988/10/17	Chrome	Linux		POINT(8.24 46.1363)	France	True	000887	2
001226	F	1954/3/29	Chrome	Windows	64 bits	POINT(113.722 34.7732)	China	False	001226	3
001247	F	1956/12/13	Safari	MacOS X		POINT(4.2591 53.3382)	Ireland	True	001247	4
001275	F	1948/5/26	Chrome	Windows	32 bits	POINT(97.822 37.751)	United States	False	001275	4
001411		1945/9/28	Firefox	Windows	32 bits	POINT(138.6005 -34.9274)	Australia	True	001411	6
001492	F	1990/6/24	Chrome	Windows	64 bits	POINT(134.044 34.3273)	Japan	True	001492	7
001571	M	1972/2/16	Chrome	MacOS X		POINT(121.23 5)	Taiwan	False	001571	3

Bước 12: Tạo đồ thị tròn



Bước 13: Tạo biểu đồ cột



Bước 14: Biểu diễn doanh số bán hàng chiến lược

